

When the vessel breaks

What to say, slide by slide

Sunday 4 October 2026 • 8 pm

🕒 17 minutes, then questions

29 slides

THE BIG IDEA

ICH is a process, not a moment.

- 1** The vessel decides where it bleeds.
- 2** The first hours decide how big it gets.
- 3** The blood keeps injuring the brain for days.

KNOW THESE BY HEART

Three lines to own

FIRST LINE	"My chapter is Chapter 5... I want to start with one person."
HEADLINE	"ICH is a process, not a moment."
LAST LINE	"Thank you. What would you like me to clarify?"

HOW TO READ EACH SLIDE

Key

KEY POINT the one idea the room must leave with

SAY talking points, with the words to land in **bold**

WORD FOR WORD learn these exactly

⏏ Pause 🖐 Ask the room 📌 Point

⦿ Prop (the golf ball) ⤵ Slow down

» the line that carries you to the next slide

Running order

	SECTION	SLIDES	TIME
00	Opening Story first. Reach Jill within 20 seconds.	Slides 1–5	2:55
01	The vessel Why a small artery gives way, and how to tell which one.	Slides 6–13	5:00
02	The first hours Rupture, growth and pressure. The golf-ball moment lives here.	Slides 14–19	3:50
03	The days after How the blood itself keeps injuring the brain.	Slides 20–24	2:55
04	At the bedside Turn the mechanisms into action, then close on Jill.	Slides 25–28	2:35
	Questions Leave the references up. Backups on slides 30 and 31.	Slide 29	Rest of slot

Three moments with the room

- **Slide 12, ~6 min:** 30 seconds with a neighbour: "Primary or secondary?"
- **Slide 16, ~9 min:** hands up for A, B or C before the dots reveal.
- **Slide 23, ~14 min:** hands up: "true or false?"
- Always **say what you see**: "Most of you went for B."
- Online: use the chat or reactions instead of hands.

The golf-ball moment

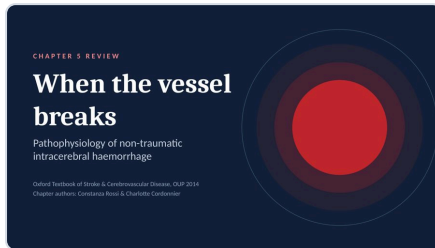
- Ball in a pocket you can reach without looking.
- On slide 18, take it out **before** you speak.
- Hold it at shoulder height. Three seconds of silence.
- Then: "Jill's surgeons removed a clot this size."
- Put it down where everyone can still see it.

Before you start

- Presenter View on; clicker tested; water within reach.
- Slides saved in three places; PDF on your phone.
- Your three messages on a card, in case the tech fails.
- Talk to two or three people before you begin.
- Feet planted, one slow breath out, find a friendly face.

If things go wrong

- **Lost your place:** read the slide title aloud and carry on.
- **Running long:** skip slides 9, 11, 15, 19 and 22 (saves about 3.5 minutes).
- **Tech fails:** give the three messages and Jill's story from your card.
- **Rambling question:** "Do you mean X or Y?"
- **Don't know:** "I don't want to guess. I'll check and come back to you."



Slide 1

⌚ 0:20

01 When the vessel breaks

🔊 WORD FOR WORD

KEY POINT

Introduce the chapter, then turn straight to one person.

SAY

- Good evening. My chapter is **Chapter 5** of the Oxford Textbook of Stroke: the pathophysiology of non-traumatic intracerebral haemorrhage.
- I want to start with **one person**.

🔊 PAUSE Walk to the centre and stand still before you say a word.

>> Advance to the date.



Slide 2

⌚ 0:35

02 10 December 1996

KEY POINT

A brain scientist wakes up with a bleed of her own.

SAY

- **10 December 1996. Seven in the morning.**
- A 37-year-old scientist wakes up with a **pounding pain behind her left eye**. She thinks it's a headache and climbs onto her exercise machine anyway.
- Her name is **Jill Bolte Taylor**. She is a neuroanatomist. She studies the brain for a living.

🔊 SLOW DOWN Tell it like a story. Slow down.

🔊 PAUSE Pause after "for a living".

>> Advance: the red dot grows.



Slide 3

🕒 0:35

03 Within four hours

KEY POINT

In four hours she lost her balance, her arm and her speech.

SAY

- Over the next **four hours**, she watches her own brain fail. First her balance, then her right arm, then her speech.
- She tries to call for help, and the numbers on the phone look like **squiggles**.
- A blood vessel had **burst** in the left side of her brain.



ASK THE ROOM Ask for a show of hands: "Who here has looked after someone with a bleed like this?" Say what you see.

>> *"Tonight is about what happened inside Jill's head, hour by hour."*



Slide 4

🕒 0:40

04 40%

KEY POINT

Four in ten die within a month, and that has not improved.

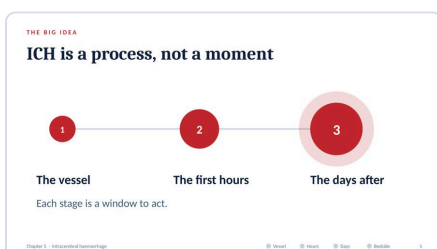
SAY

- **Forty percent.** That is how many people die within a month of an intracerebral haemorrhage.
- That number **did not move between 1980 and 2008**. In the same years, treatment for ischaemic stroke was transformed.
- Part of the problem: we treat a bleed as **one moment**. The vessel bursts, the damage is done, and we wait.



POINT Point at the number and stay silent for two seconds.

>> *"Chapter 5 gives us a different view."*



Slide 5

🕒 0:45

05 ICH is a process, not a moment

🗨️ WORD FOR WORD

KEY POINT

The headline of the whole talk.

SAY

- **ICH is a process, not a moment.**
- The **vessel** decides where it bleeds. The **first hours** decide how big it gets. Over the following **days**, the blood itself keeps injuring the brain.
- Watch the red circles grow. That is the bleed getting bigger.
- Each stage is a **window where we can act**.



PAUSE Say the headline slowly, then pause for two seconds.

>> *"We'll take them one at a time."*

01 The vessel

Why a small artery gives way, and how to tell which one.

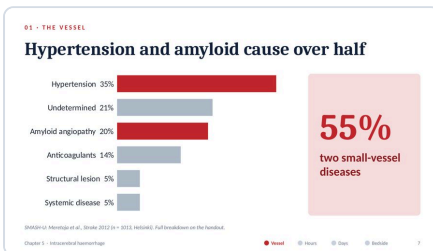
⌚ 5:00



Slide 6

⌚ 0:05

- "Stage one. **The vessel**. Why does a small artery give way?"



Slide 7

⌚ 0:45

07 Hypertension and amyloid cause over half

KEY POINT

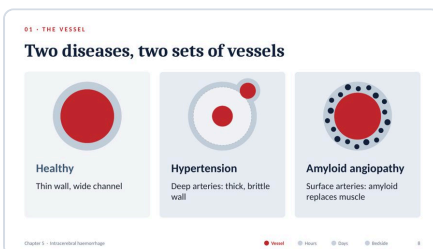
Two small-vessel diseases cause more than half of all bleeds.

SAY

- This is **SMASH-U**: just over a thousand patients in Helsinki, each sorted by cause.
- Hypertension explained **35 percent**. Amyloid angiopathy explained **20**. Together that is more than half, and both are diseases of small vessels.
- Look at the second bar: **one in five** had no clear cause, even after full work-up.

➤ **POINT** Point at the two red bars and stay silent for two seconds.

» "So what do these two diseases do to a small artery?"



Slide 8

⌚ 1:00

08 Two diseases, two sets of vessels

KEY POINT

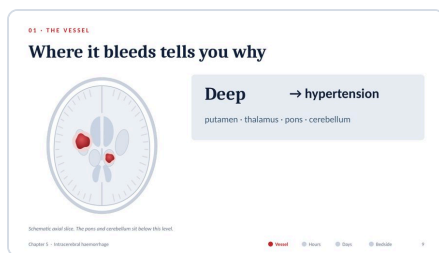
Hypertension damages deep arteries; amyloid damages surface arteries.

SAY

- Left: a **healthy** artery. Thin wall, wide channel.
- Middle: **hypertension**. It attacks the tiny perforating arteries deep in the brain. Years of high pressure make the wall thick and glassy, but **brittle**.
- The bulge is a **Charcot-Bouchard microaneurysm**, first described in 1868. Its role in rupture is still debated.
- Right: **amyloid angiopathy**. The dark dots are beta-amyloid. It replaces the muscle in arteries on the surface of the brain. Mostly over 55.

➤ **POINT** Walk them left to right: say what it is, point at what matters, then say what it means.

» "That gives us a rule you can use tomorrow."



Slide 9

🕒 0:30

09 Where it bleeds: deep

KEY POINT

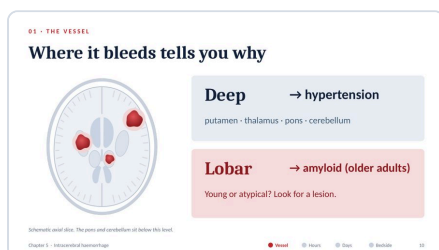
Deep bleed: think hypertension.

SAY

- **Where it bleeds tells you why.**
- These two are **deep**: the putamen and the thalamus. The pons and cerebellum count too.
- The perforating arteries branch at **right angles** off big, high-pressure trunks, so they take the full force of the blood pressure.

➤ **POINT** Point to the two deep bleeds.

» *Advance: a third bleed appears near the surface.*



Slide 10

🕒 0:30

10 Where it bleeds: lobar

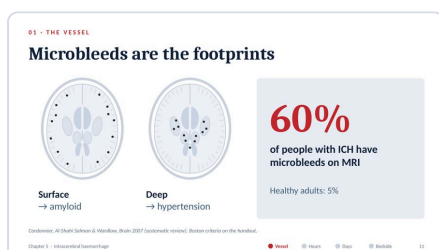
KEY POINT

Lobar bleed in an older adult: think amyloid. Young or atypical: be suspicious.

SAY

- This one is **lobar**: in the cortex and the white matter just beneath it.
- In an **older adult**, a lobar bleed points to amyloid angiopathy.
- In a **younger** person, or when something looks unusual, it should make us **suspicious**.

» *"Keep that thought. We'll come back to Jill in a moment."*



Slide 11

🕒 0:40

11 Microbleeds are the footprints

KEY POINT

Microbleeds on MRI reveal which disease is at work.

SAY

- Microbleeds are tiny old leaks of blood. On MRI they show up as **black dots**.
- Dots near the **surface** point to amyloid. Dots **deep** inside point to hypertension.
- Charlotte Cordonnier, **one of this chapter's authors**, found them in **60 percent** of people with ICH, against 5 percent of healthy adults.
- The handout has the Boston criteria, which use these dots to diagnose amyloid in life.

» *"Your turn."*

YOUR TURN - 30 SECONDS WITH YOUR NEIGHBOUR

Jill: primary or secondary?

37 years old No hypertension Lobar bleed

?

Chapter 5 - Intracerebral haemorrhage

Slide 12

⌚ 0:45

12 Jill: primary or secondary?

KEY POINT

Let the room work out Jill's diagnosis. This is soft break 1, at about minute 6.

SAY

- Jill was 37. She had **no history of high blood pressure**. Her bleed was **lobar**.
- Primary small-vessel disease, or a secondary cause? **Turn to the person next to you. Thirty seconds.**

👉 **ASK THE ROOM** Stop talking. Count to 30 in your head, then ask two people for their answer.

>> *Advance: "Secondary."*

YOUR TURN - 30 SECONDS WITH YOUR NEIGHBOUR

Secondary: an arteriovenous malformation

37 years old No hypertension Lobar bleed

65%
of patients like Jill had a lesion on angiography

0%
if over 45, hypertensive, with a deep bleed

Zhu et al., Stroke 2017 (prospective, 200 patients, all had catheter angiography). Other secondary causes on the timeline.

Chapter 5 - Intracerebral haemorrhage

Slide 13

⌚ 0:45

13 Secondary: an AVM

KEY POINT

Jill's profile was a textbook warning sign for a hidden lesion.

SAY

- Secondary.** Jill was born with an **arteriovenous malformation**, a tangle of vessels with no capillaries in between. That morning it ruptured.
- In 206 patients who all had angiography, **65 percent** of young, normotensive patients with a lobar bleed had a vascular lesion.
- Over 45, hypertensive, with a deep bleed: **zero**.
- Same test, completely different result. **The patient's story tells you where to look.**

>> *"Stage two."*

02 The first hours

Rupture, growth and pressure. The golf-ball moment lives here.

⌚ 3:50

02

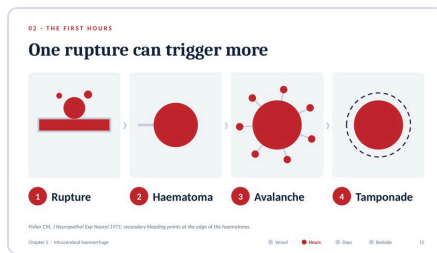
The first hours

Rupture, growth and pressure

Slide 14

⌚ 0:05

- "Stage two. **The first hours.** Jill had four of them. Let's see what happens in that time."



Slide 15

🕒 0:45

15 One rupture can trigger more

KEY POINT

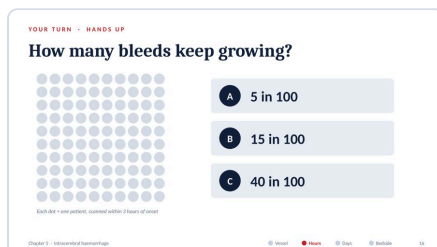
A bleed keeps spreading by tearing the vessels around it.

SAY

- **One:** a diseased vessel ruptures, with arterial pressure behind it.
- **Two:** blood splits the tissue and a haematoma forms.
- **Three** is the surprise. In 1971 Miller Fisher found extra bleeding points around the edge. The growing clot tears the vessels around it: an **avalanche**.
- **Four:** the bleeding stops when tissue pressure and clotting finally win.

➤ **POINT** Move left to right across the four pictures.

» "So how often does the bleeding keep going?"



Slide 16

🕒 0:30

16 How many bleeds keep growing?

KEY POINT

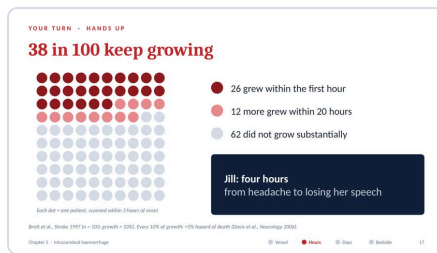
Get the room to commit to a guess before the reveal. This is soft break 2, at about minute 9.

SAY

- Here are **100 patients**, all scanned within three hours of onset.
- How many of their bleeds keep growing? **A, five. B, fifteen. C, forty.**

👉 **ASK THE ROOM** Ask for hands up for each option. Say what you see: "Most of you went for B."

» *Advance: the dots colour in.*



Slide 17

🕒 0:45

17 38 in 100 keep growing

KEY POINT

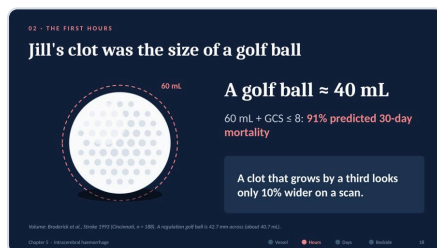
38 in 100 bleeds keep growing, and growth predicts death.

SAY

- **Thirty-eight.**
- **26** grew by more than a third within one hour of the first scan. **12 more** within twenty hours. The growth went with neurological decline.
- This is **Jill's four hours**. The damage kept spreading because the bleed kept growing.
- Every 10 percent of growth raised the risk of death by **5 percent**. Highest risk: scanned early, large bleeds, anticoagulants, a spot sign on CT angiography.

🔊 **PAUSE** Wait for the dots to finish colouring before you speak.

>> "Jill's surgeons found out how big hers got."



Slide 18

🕒 1:00

18 Jill's clot was the size of a golf ball

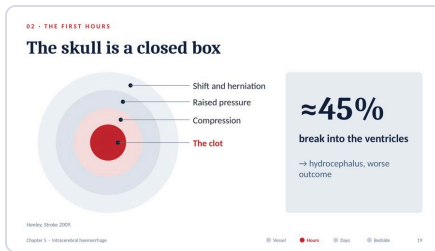
KEY POINT

Small changes in size are deadly, and easy to miss. This is your jaw-dropping moment.

SAY

- Jill's surgeons removed a clot **this size**. A golf ball.
- A golf ball holds about **40 millilitres**. Grow it to 60, the dashed line. It barely looks bigger, does it?
- A clot that grows by a third looks only **10 percent wider** on a scan. Easy to miss by eye. The patient feels every millilitre.
- At 60 millilitres with a GCS of 8 or less, predicted 30-day mortality was **91 percent**.
- **PROP** Take the golf ball out BEFORE you speak. Hold it at shoulder height and stay silent for three seconds. Put it down where everyone can still see it.

>> "Why does volume matter so much?"



Slide 19

🕒 0:45

19 The skull is a closed box

KEY POINT

Inside a closed skull, extra volume becomes extra pressure.

SAY

- Because **the skull is a closed box**.
- The clot compresses the tissue around it. As the volume grows, **pressure rises**. If it keeps rising, the brain shifts and can **herniate**.
- About **45 percent** of bleeds break into the ventricles. Blood there can block CSF flow and cause hydrocephalus.

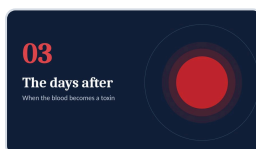
➤ **POINT** Trace the rings from the centre outwards.

» "Stage three."

03 The days after

How the blood itself keeps injuring the brain.

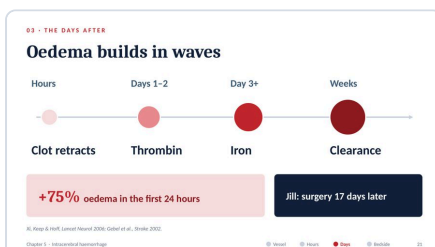
🕒 2:55



Slide 20

🕒 0:05

- "Stage three. **The bleeding has stopped. The injury has not.**"



Slide 21

🕒 0:55

21 Oedema builds in waves

KEY POINT

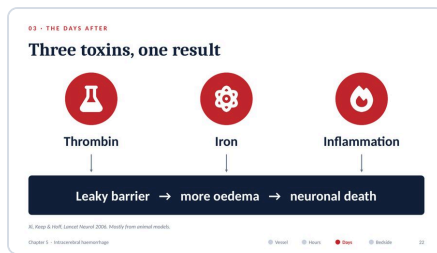
Swelling builds in waves for up to two weeks.

SAY

- **Hours:** the clot retracts, like a blood sample settling in a tube, and squeezes serum into the tissue.
- **Days 1-2:** the clotting cascade makes thrombin, which opens the blood-brain barrier.
- **Day 3 on:** red cells break down and release iron. **Weeks:** scavenger cells clear the clot and leave a cavity.
- Oedema grew by **75 percent** in the first 24 hours. Jill's surgery came **17 days later**, and the clot was still pressing on her language centres.

➤ **POINT** Move along the timeline, left to right.

» "Three toxins drive this second wave of injury."



Slide 22

🕒 0:45

22 Three toxins, one result

KEY POINT

Thrombin, iron and inflammation keep injuring the brain after the bleeding stops.

SAY

- **Thrombin** comes first. We need it to stop the bleed, but at high levels it damages the barrier and kills cells.
- **Iron** comes later. Released from burst red cells, it fuels free radicals.
- **Inflammation** runs throughout. Immune cells arrive within hours and stay for days.
- All three end in the same place: a leaky barrier, more swelling, dead neurons. **That is why patients get worse days later.**

👉 **POINT** Point to the dark bar at the end.

>> "True or false?"



Slide 23

🕒 0:30

23 True or false?

KEY POINT

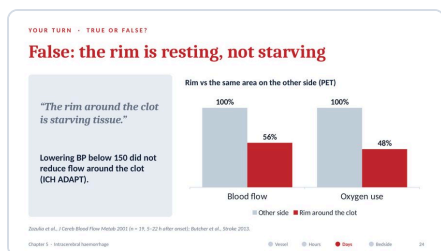
Test a belief many clinicians hold. This is soft break 3, at about minute 14.

SAY

- "The dark rim around the clot is **starving tissue**, like the penumbra in an ischaemic stroke." **Hands up for true.**
- This matters, because for years people worried that lowering blood pressure would starve that rim.

👉 **ASK THE ROOM** Count the hands and say what you see.

>> *Advance: "False."*



Slide 24

⌚ 0:40

24 False: the rim is resting, not starving

KEY POINT

The rim around the clot is resting, not starving.

SAY

- **False.** PET in 19 patients, 5 to 22 hours after onset.
- Blood flow in the rim was **56 percent** of the other side. Oxygen use was even lower: **48 percent**. The tissue is **resting, not starving**.
- In ICH ADAPT, lowering systolic pressure **below 150** did not reduce flow around the clot.

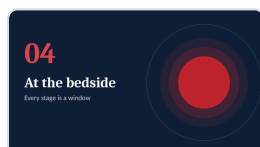
➤ **POINT** Point to each pair of bars in turn.

» "So what do we do with all this?"

04 At the bedside

Turn the mechanisms into action, then close on Jill.

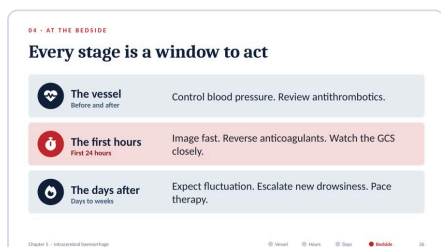
⌚ 2:35



Slide 25

⌚ 0:05

- "So what do we do with all this **at the bedside**?"



Slide 26

⌚ 1:15

26 Every stage is a window to act

KEY POINT

Every stage of the process gives us something to do. This is the heart of the talk.

SAY

- **The vessel:** blood pressure control is our strongest tool against a first or second bleed. Review antithrombotics.
- **The first hours:** image fast, lower blood pressure early, reverse anticoagulants urgently. Watch the GCS closely. **A falling GCS is not something to watch overnight.**
- **The days after:** expect fluctuation as swelling peaks. New drowsiness, headache or vomiting means **stop and escalate**. Reassess before every session and pace therapy.

➤ **SLOW DOWN** Slow right down here. Give each row its own breath.

» "If you remember three things from tonight..."

SUMMARY

If you remember three things

1

The vessel decides where it bleeds.

2

The first hours decide how big it gets.

3

The blood keeps injuring the brain for days.

Chapter 5 · Introduction to Neuroscience

Vessel

Hours

Days

Stroke

27

Slide 27

🕒 0:30

27 If you remember three things

🔊 WORD FOR WORD

KEY POINT

The three messages, one last time.

SAY

- The **vessel** decides where it bleeds.
- The **first hours** decide how big it gets.
- The **blood** keeps injuring the brain for days.

🔊 **PAUSE** Pause between each line.

» "Back to Jill."

EIGHT YEARS LATER

Jill walked onto the TED stage holding a human brain.

ICH is a process, not a moment.

What would you like me to clarify?

Slide 28

🕒 0:45

28 Eight years later

🔊 WORD FOR WORD

KEY POINT

Close the story and land the big idea.

SAY

- Back to Jill. It took her **eight years** to recover. Then, in 2008, she walked onto the TED stage holding a real human brain.
- Her bleed was not a moment. It was a process, and so was her recovery.
- **ICH is a process, not a moment.**
- Thank you. **What would you like me to clarify?**

🔊 **PAUSE** Pause for three full seconds after the big idea. If the room is quiet, ask the discussion question and wait five seconds.

» Questions.



Questions

Leave the references up while you take questions.

SOURCES

References

1. Bandt C, Condon C. Pathophysiology of non-traumatic intracerebral hemorrhage. In: Hemming R, ed. Oxford Textbook of Stroke and Cerebrovascular Disease. Oxford: Oxford University Press; 2016:46-53.
2. Taylor JE. My stroke of insight. TED 2008, and My Stroke of insight (book); 2008.
3. van den Broek C, et al. Lancet Neurol 2010;9:167-176.
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6. Zhu XL, Chen H, Poon W, et al. Stroke 1997;28:1458-1469.
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8. Bandt C, et al. Stroke 1997;28:1-5.
9. Davis SM, et al. Neurology 2006;66:1175-1180.
10. Broderick JP, et al. Stroke 1993;24:967-973.
11. Hemming R. Stroke 2009;40:1510-1516.
12. Fisher CM, et al. Stroke 2003;34:2011-2015.
13. Xie G, et al. Lancet Neurol 2006;5:51-63.
14. Zandi PP, et al. Lancet Neurol 2005;4:404-410.
15. Broderick JP, et al. JAMA 2001;286:1632-1636.

Slide 29

29 References

KEY POINT

Leave this up while you take questions.

SAY

- Every figure in the talk comes from these papers. **The first reference is the chapter itself.**
- Hidden backups: type **30** and press Enter for other secondary causes, or **31** for what has changed since 2014.



Likely questions

Answer in under 30 seconds: answer, one fact, stop.

Why do deep perforators bleed, not big arteries?

They branch at right angles off high-pressure trunks, with little buffering. Hypertension leaves the wall thick but brittle.

Are Charcot-Bouchard microaneurysms real?

They were described in 1868, but some later proved to be artefacts. Their role in rupture is still debated.

How do you confirm amyloid angiopathy?

Definite CAA needs pathology. In life we use the Boston criteria (lobar microbleeds, superficial siderosis), and v2.0 (2022) added white-matter markers.

Why are anticoagulant bleeds so bad?

They keep growing for longer. In SMASH-U they were the largest bleeds, with 54% dead at three months. Reverse urgently.

Is lowering BP risky for the rim?

No. PET showed the rim is resting, not starving. ICH ADAPT found BP below 150 did not reduce flow, and INTERACT2 found below 140 safe.

What is the spot sign?

Contrast leaking into the clot on CT angiography. It means active bleeding and predicts growth.

How is haemorrhagic transformation different?

The infarct comes first, then the damaged tissue bleeds, often after reperfusion.

Why do patients worsen on days 3–5?

Rising oedema, iron release, hydrocephalus, plus pneumonia or seizures.

Is there a drug for secondary injury?

Not yet. The i-DEF trial of deferoxamine (2019) showed no meaningful benefit.

When should rehab start?

Follow the team's parameters and watch closely for the first 24–72 h. AVERT found very early, high-dose mobilisation within 24 h reduced good outcomes. Use short, frequent sessions once stable.

Is ICH more common in some settings?

Yes. It is about 28% of strokes worldwide and highest in lower-middle-income countries. Hypertension drives it.

A question you can't answer

"Good question. I don't want to guess. I'll check and come back to you."

If the room goes quiet

- Ask: "Which of these windows do we meet most often in our own practice, and are we timing our assessments around it?"
- Then wait a full five seconds. Silence gives people time to think.